Changepoint detection for fractional Brownian motion

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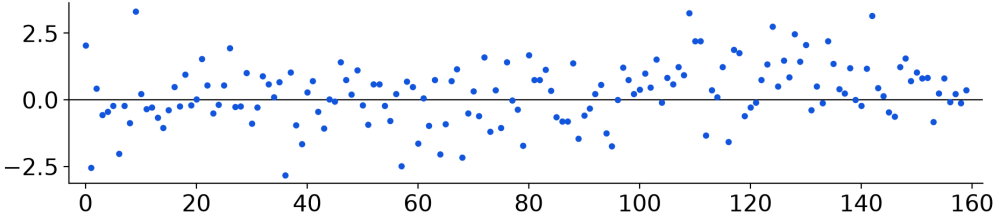
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Introduction

A changepoint is an unknown moment of time at which the probabilistic characteristics of an observed process change.

The problem of sequential changepoint detection is to detect the change in real time.

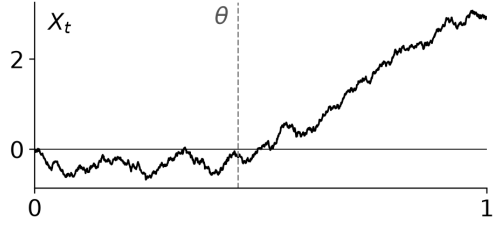


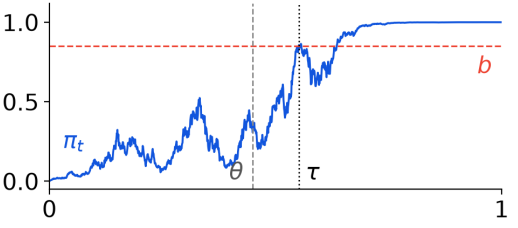
The classical problem in continuous time

Suppose a drift appears in a Brownian motion at a random time θ, so we observe

$$X_t = \mu(t - \theta)^+ + B_t.$$

Solution (Shiryaev, 1963): if θ has an exponential distribution, then it is optimal to stop when the posterior probability $\pi_t = \mathrm{P}(\theta \le t \mid \mathcal{F}_t^X)$ exceeds a threshold.





Solution of the classical problem

For the criterion $\mathrm{E}(\tau-\theta)^+ + c\,\mathrm{P}(\tau<\theta) \to \min$, the detection problem reduces to the Markovian optimal stopping problem

$$\inf_{\tau}\ \mathrm{E}\left(\int_0^{\tau} \pi_s\,ds + c\,(1-\pi_\tau)\right),$$

where the posterior probability process π is a one-dimensional homogeneous diffusion

$$d\pi_t = \lambda(1-\pi_t)\,dt + \mu\,\pi_t(1-\pi_t)\,d\bar{B}_t.$$

The problem is solved through a free-boundary problem for an ODE.

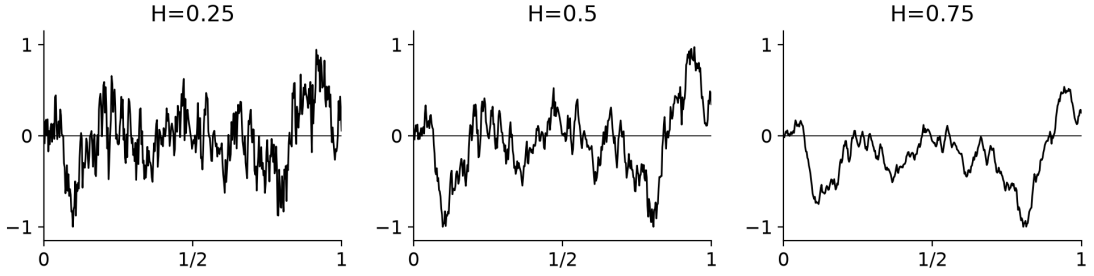
Changepoint in a fractional Brownian motion

Fractional Brownian motion $(B_t^H)_{t\geq 0}$ with Hurst parameter $H \in (0,1)$ is a zero-mean continuous Gaussian process with covariance function

$$\mathrm{E}(B_t^H B_s^H) = \frac{1}{2}\left(t^{2H} + s^{2H} - |t-s|^{2H}\right).$$

- If $H = \frac{1}{2}$, then B^H is a standard Brownian motion.
- If $H > \frac{1}{2}$ (resp. $H < \frac{1}{2}$), the increments of B^H are positively (negatively) correlated.
- Widely used for modeling in finance, telecommunications, hydrology, etc.
- Not a Markov process, not a semimartingale.

Sample paths of fractional Brownian motion



The changepoint detection problem for fBm

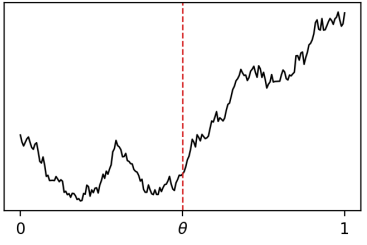
We observe the process

$$X_t = \mu(t-\theta)^+ + B_t^H,$$

where

– θ is the (unobservable) changepoint,

– $\mu > 0$ is a known drift.



Bayesian setting:

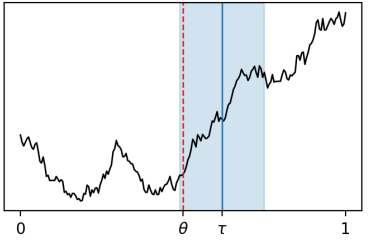
$$\theta \text{ is independent of } B^H, \qquad \theta \sim U([0,T]).$$

A detection rule is a stopping time $\tau \leq T$ of the filtration $\mathcal{F}_t^X = \sigma(X_s,\, s \leq t)$. The set of all such rules is denoted by $\mathcal{S}$.

Optimality criterion

We call $\tau^* \in \mathcal{S}$ h-optimal for $h \in (0, T/2)$ if

$$\mathrm{P}\big(|\theta - \tau^*| < h\big) = \sup_{\tau \in \mathcal{S}} \mathrm{P}\big(|\theta - \tau| < h\big).$$



Our main result is a numerical method of finding an h-optimal detection rule.

Reduction to an optimal stopping problem

Step 1: from fBm to standard Brownian motion

FBm can be transformed into a standard Brownian motion by the formula (Molchan, Golosov (1969), Jost (2006))

$$B_t = \int_0^t K(t,s)\, dB_s^H$$

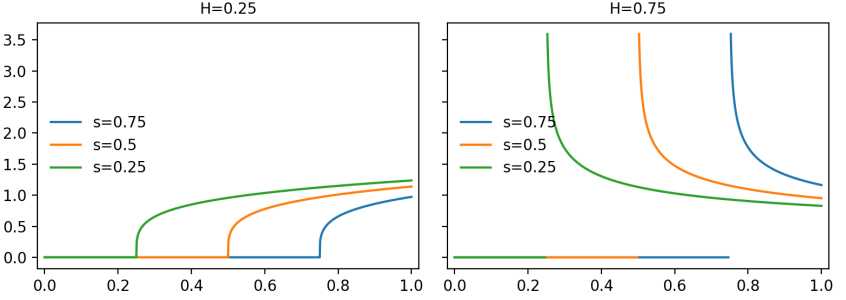
with the kernel $K(t,s)$ known in a closed form.

Applying the same transformation to the observed process X, introduce the new process

$$Y_t = \int_0^t K(t,s)\, dX_s = \int_0^t \mu(u,\theta)\, du + B_t,$$

which is a changepoint problem for standard BM, but with a nonlinear drift $\mu(\cdot\,,\theta)$.

The drift function $\mu(u, s)$:



Step 2: reduction to an optimal stopping problem

Let $\widetilde{\mathrm{P}}$ be the probability measure under which no change in the drift occurs, so that Y is a standard Brownian motion under $\widetilde{\mathrm{P}}$.

For the measure $\mathrm{P}^s(\cdot) = \mathrm{P}(\cdot \mid \theta = s)$, by Girsanov's theorem, we have

$$L_t^s := \frac{d\mathrm{P}^s \mid \mathcal{F}_t^X}{d\widetilde{\mathrm{P}} \mid \mathcal{F}_t^X} = \exp\left(\int_s^t \mu(u,s)\, dY_u - \frac{1}{2}\int_s^t \mu^2(u,s)\, du\right), \qquad t > s.$$

Define the likelihood ratio moving average

$$A_t = \frac{1}{h}\int_{t-h}^t L_t^s\, ds.$$

Theorem. The changepoint detection problem is equivalent to the optimal stopping problem

$$V = \sup_{h \le \tau \le T-h} \widetilde{\mathrm{E}} A_\tau.$$

In this problem, an optimal τ^* exists, and it is the optimal changepoint detection rule:

$$\sup_{\tau \in \mathcal{S}} \mathrm{P}(|\theta - \tau| < h) = \mathrm{P}(|\theta - \tau^*| < h) = \frac{h}{T}(1 + V).$$

Numerical solution

Discretization

Let $\Delta t = T/N$ and assume that an interval $[t-h, t]$ contains $d = h/\Delta t$ steps.

Replacing the integrals in L and A by sums, we get the optimal stopping problem

$$V^N = \sup_{\nu \in \mathcal{S}^N} \widetilde{\mathrm{E}} A^N_\nu, \qquad A^N_n = \frac{1}{d} \sum_{k=n-d}^{n-1} e^{R^k_n},$$

where R^k_n are Gaussian variables built from ΔY_{t_m}, $k < m \le n$, and $\mathcal{S}^N$ is the set of stopping times taking values in $\{d, \dots, N-d\}$.

This is a d-dimensional Markov optimal stopping problem with the state vector

$$Z_n = (\Delta Y_{t_{n-d+1}}, \dots, \Delta Y_{t_n}) \in \mathbb{R}^d.$$

Proposition. Let $\gamma = \min\left(\frac{1}{2}, 1 - H\right)$. Then

$$|V - V^N| = O\left(\frac{1}{N^\gamma}\right).$$

Neural network approach

We bound the value V^N from both sides:

$$\widehat{V}^N_{\text{primal}} \leq V^N \leq \widehat{V}^N_{\text{dual}}.$$

- In the primal problem, we optimize over stopping rules.
- In the dual problem, we optimize over martingales.

The primal problem

– We consider randomized stopping times, i.e. stop at step n with probability

$$\lambda_n = \lambda(n, Z_n) \in [0, 1].$$

– The function λ is approximated by a neural network $\lambda(n, z; p)$.

– $\widetilde{\mathrm{E}} A_\lambda^N$ is evaluated by the Monte Carlo method and a stochastic gradient ascent method is used to find the best parameters p^*.

The dual problem

Proposition (Rogers, 2002). The following representation holds true:

$$V^N = \inf_{M \in \mathcal{M}^N} \widetilde{\mathrm{E}} \max_{d \le n \le N-d} (A_n^N - M_n),$$

where $\mathcal{M}^N$ is the set of martingales with $M_d = 0$.

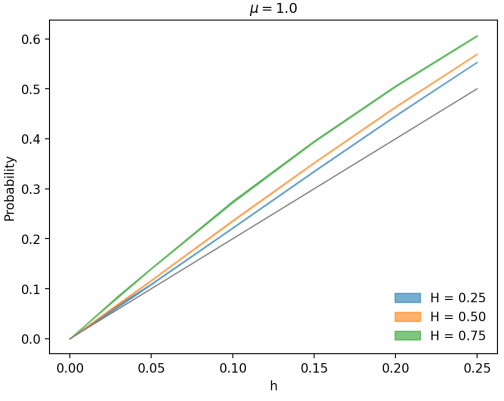
The increments of the optimal martingale admit the Hermite expansion

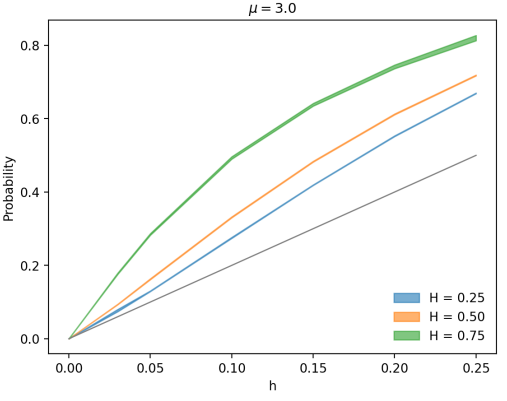
$$\Delta M_n^* = \sum_{i=1}^{\infty} h^i(n, Z_{n-1})\, H_i\left(\frac{\Delta Y_{t_n}}{\sqrt{\Delta t}}\right) \quad \text{with some functions } h^i(n,z).$$

We truncate the expansion at q terms and approximate $h^1, \dots, h^q$ by another neural network, trained to minimize the upper bound.

Examples

Probability of covering the changepoint by the optimal interval ($T = 1$, $N = 100$).





Thank you for your attention